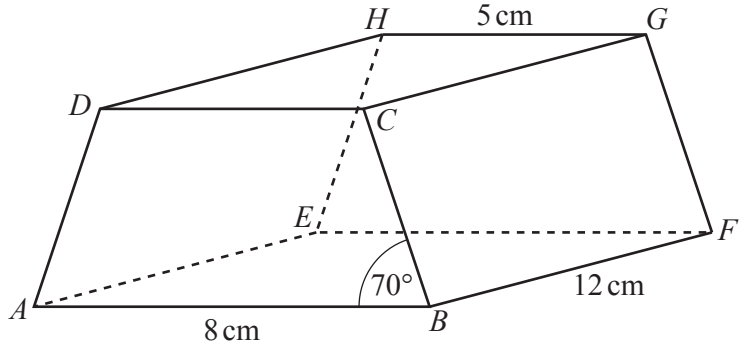
0580_w20_qp_42_Q9

9



NOT TO SCALE

The diagram shows a prism with a rectangular base, *ABFE*.
The cross-section, *ABCD*, is a trapezium with $AD = BC$.
$AB = 8$ cm, $GH = 5$ cm, $BF = 12$ cm and angle $ABC = 70°$.

(a) Calculate the total surface area of the prism.

..... cm^2 [6]

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(b) The perpendicular from G onto EF meets EF at X.

(i) Show that $EX = 6.5$ cm.

[1]

(ii) Calculate AX.

$AX =$ ..... cm [2]

(iii) Calculate the angle between the diagonal AG and the base $ABFE$.

..... [2]

0580_w20_ms_42_Q9

Question	Answer	Marks	Partial Marks
9(a)	315 or 314.5 to 315.0	**6**	**M1** for tan70 = $\frac{\text{height}}{\frac{1}{2}(8-5)}$ oe or better seen **M1dep** for $\frac{1}{2}(8+5)\times$ *their* height or better seen **dep** on trig attempt for height **M2** for 12 × $\frac{\frac{1}{2}(8-5)}{\cos 70}$ oe or better seen or **M1** for $\frac{\frac{1}{2}(8-5)}{\cos 70}$ oe or better seen **M1** for 8 × 12 oe isw and 5 × 12 oe isw
9(b)(i)	8 – ½ (8 – 5) or 5 + ½ (8 – 5)	**M1**	
9(b)(ii)	13.6 or 13.64 to 13.65	**2**	**M1** for $12^2 + (6.5)^2$ oe
9(b)(iii)	16.8 or 16.9 or 16.79 to 16.91… nfww	**2**	**M1** for identifying angle *GAX* from a diagram or from working or better